

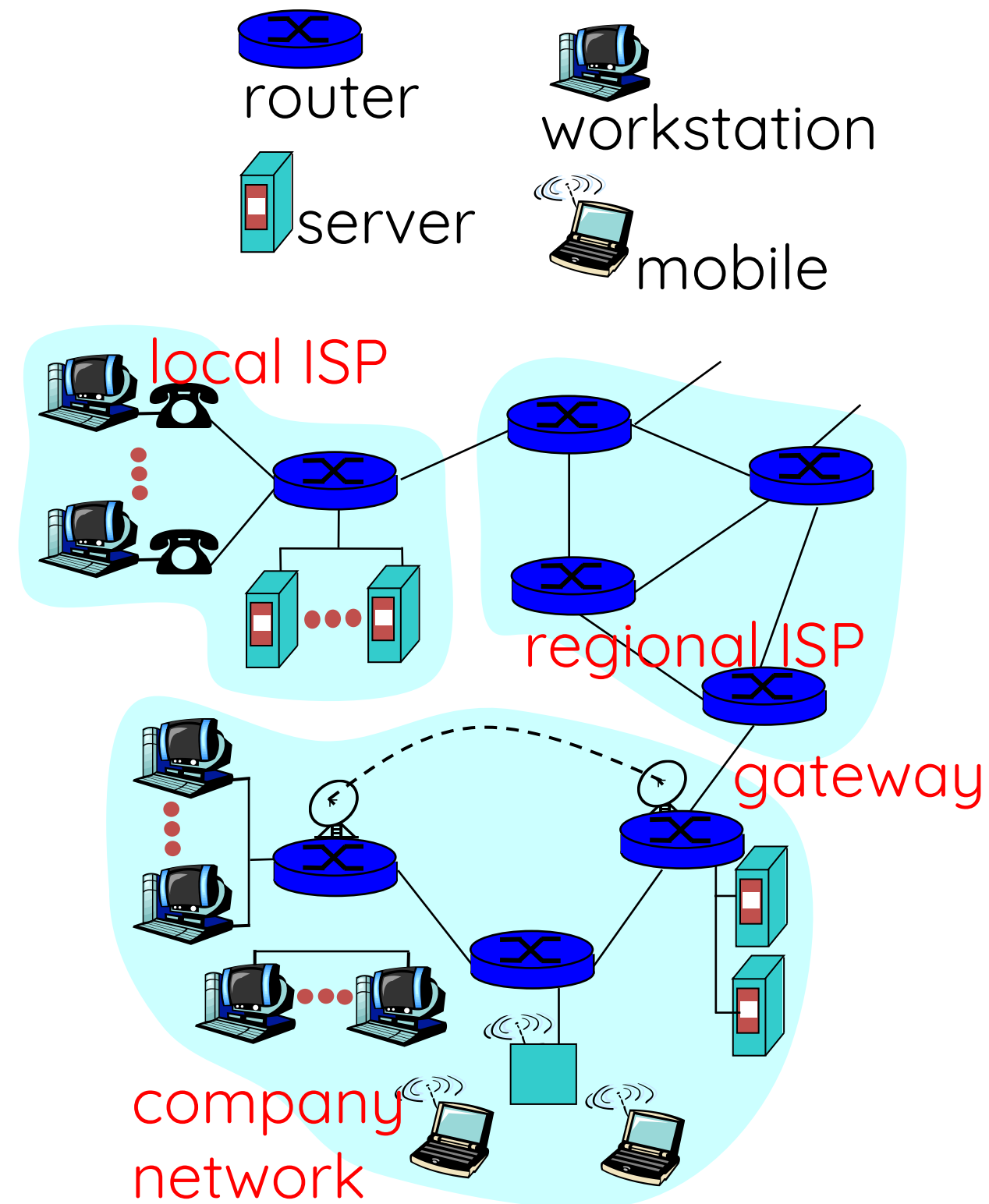
Design of the Internet

Module 1.2 – Switching and Protocols

Learning Goals

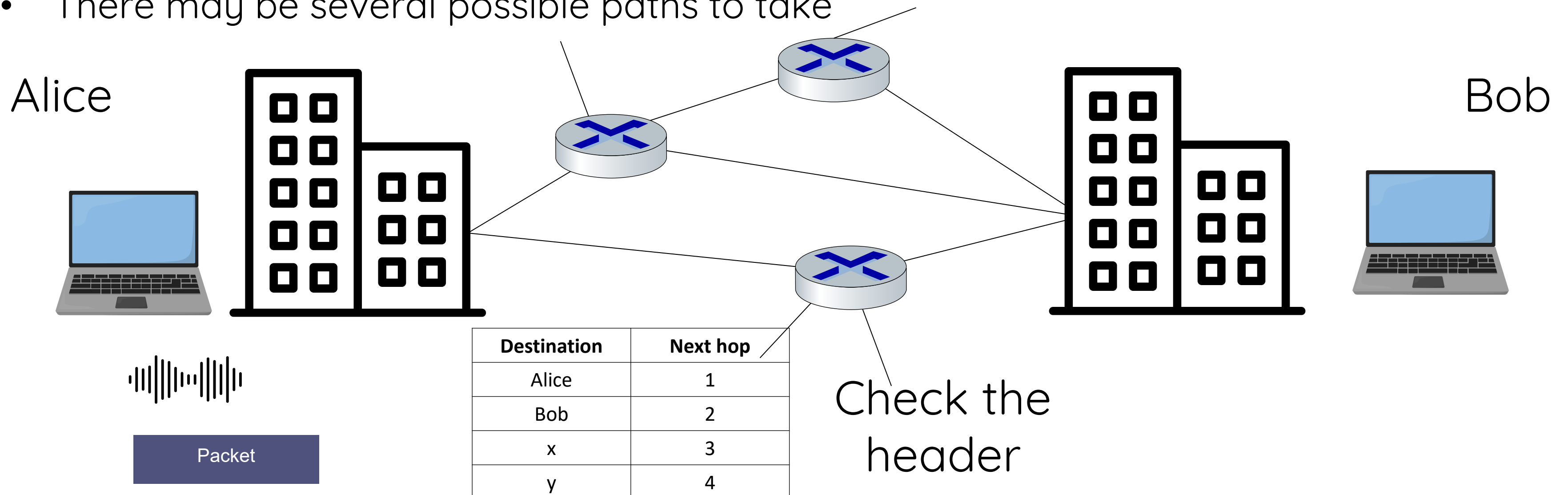
- A. Explain what the Internet is, and how (generally) it is put together
- B. Describe the advantages and disadvantages of packet vs circuit switching
- C. Define the term protocol in the context of a computer network
- D. Explain why protocols are modeled using finite state machines and what states, actions, and events are
- E. List the layers of the Internet protocol stack, in their proper order, and what functions each layer performs
- F. Explain encapsulation and the structure of packets flowing in the Internet

As a network of networks, the internet has many routers



Packet Switching and Routers

- Packet is sent to the router that is believed to be closest to the destination
- Router looks up destination address in a forwarding table to determine next hop
- There may be several possible paths to take



Circuit vs packet switching revisited

- When using circuit switching, the decision about the route that the data will take from source to destination is made once when the connection is established
- When using packet switching, this decision is made for every packet
- Does this have any impact on reliability?
- Does this have any impact on performance?

Clicker Question

- Assume A creates a 100kbps circuit to B, and A sends data at an average rate of 25kbps.

What is A's utilization of the network resources? Express your answer as a percent.

Clicker Answer

- Assume A creates a 100kbps circuit to B, and A sends data at an average rate of 25kbps.

What is A's utilization of the network resources? Express your answer as a percent.

$$25\text{kbps} / 100\text{kbps} = 0.25 \text{ or } 25\%$$

Clicker Question

- Assume A creates a 100kbps circuit to B, and A sends data at an average rate of 25kbps.
- Can the network use these idle resources for other traffic?
 - A. Yes
 - B. No

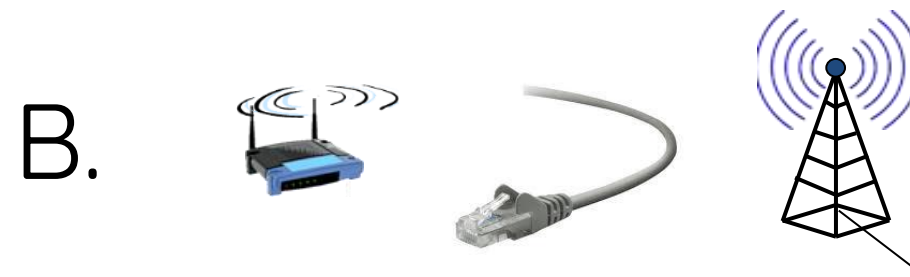
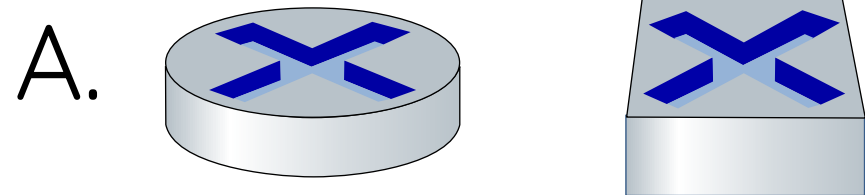
Clicker Question

- Assume A creates a 100kbps circuit to B, and A sends data at an average rate of 25kbps.
- Can the network use these idle resources for other traffic?
 - A. Yes
 - **B. No**

Those resources are committed to that circuit and must be available to it if they are needed.

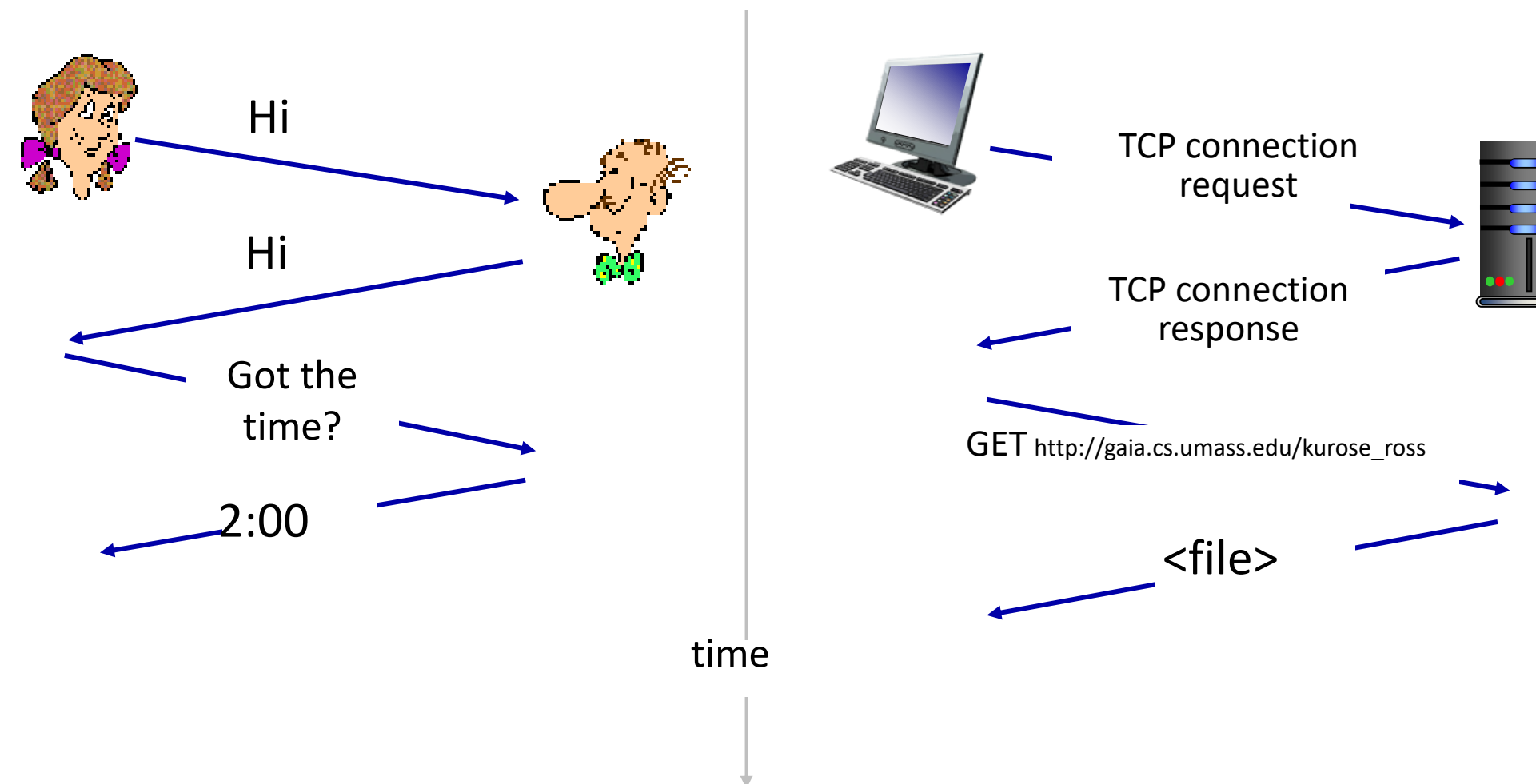
IClicker Game

A-B-C game:



What's a protocol?

A human protocol and A computer network protocol:

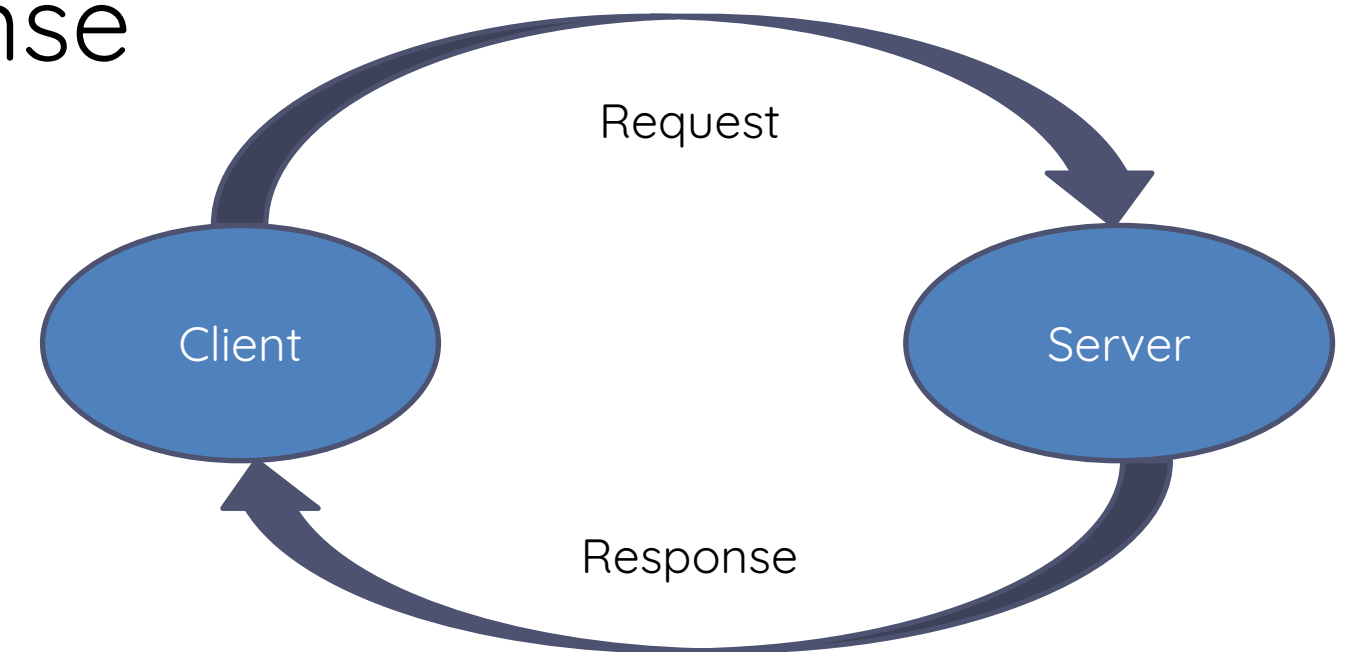


Protocols

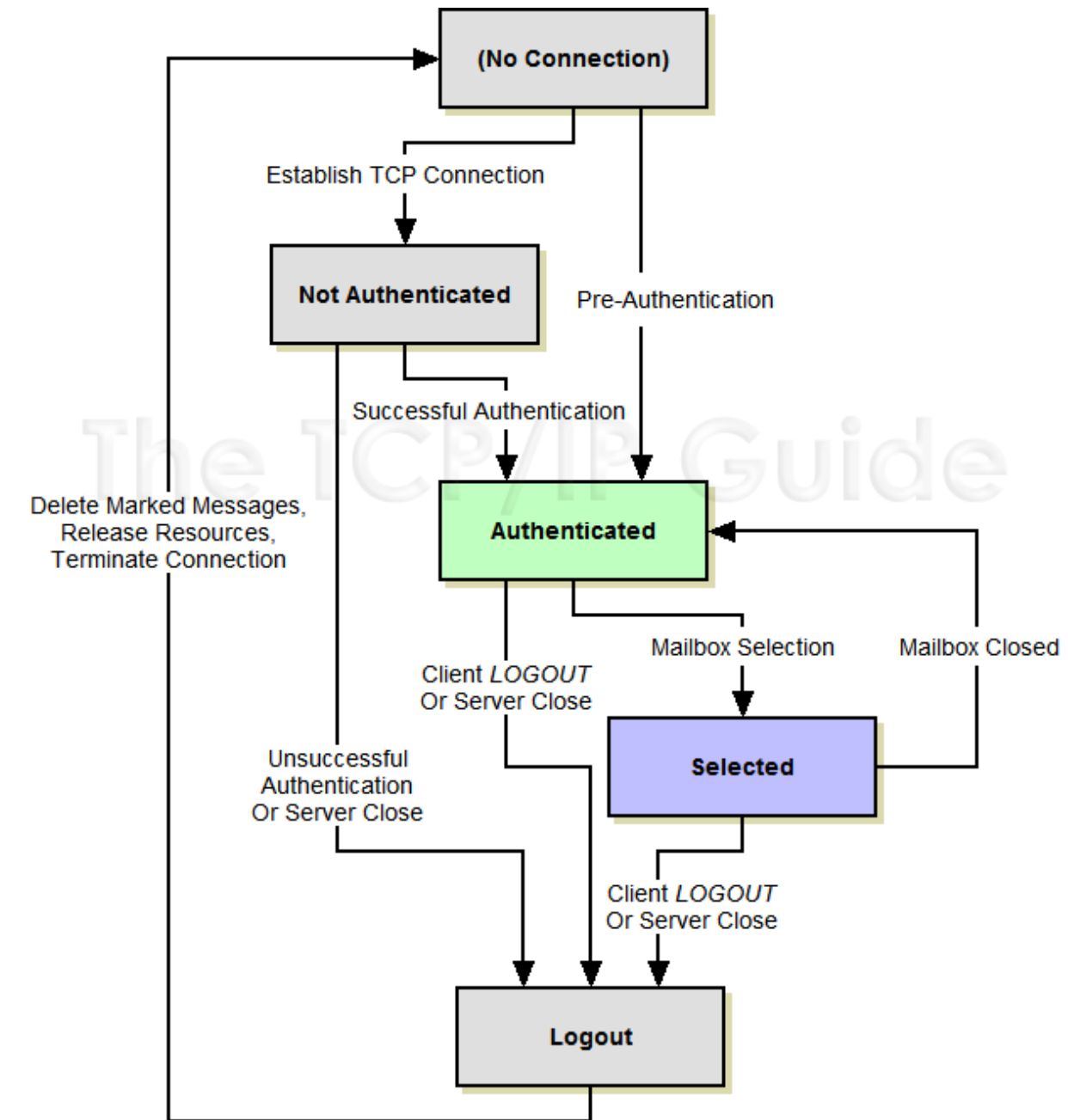
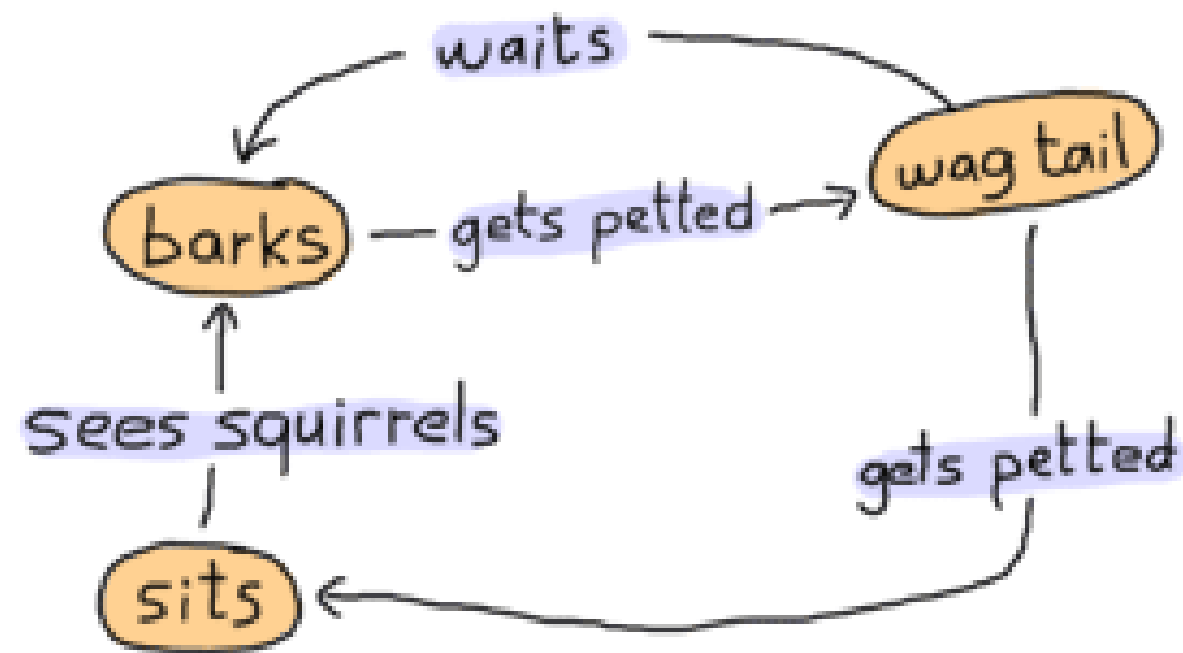
- A protocol defines:
 - Roles of communicating entities
 - Format of messages
 - Order of messages
 - Actions taken on the transmission, receipt of a message, or other event
- A fully-defined protocol must provide a proper action for any event in any state

Request-Response Protocols

- Many protocols on the Internet are request-response protocols
 - Requestor (usually client) sends a request
 - Receiver (usually server) sends a response
 - Well-defined rules for whose turn it is
- Some rules can be complicated
 - Server is slow to respond
 - Size of request or response can vary

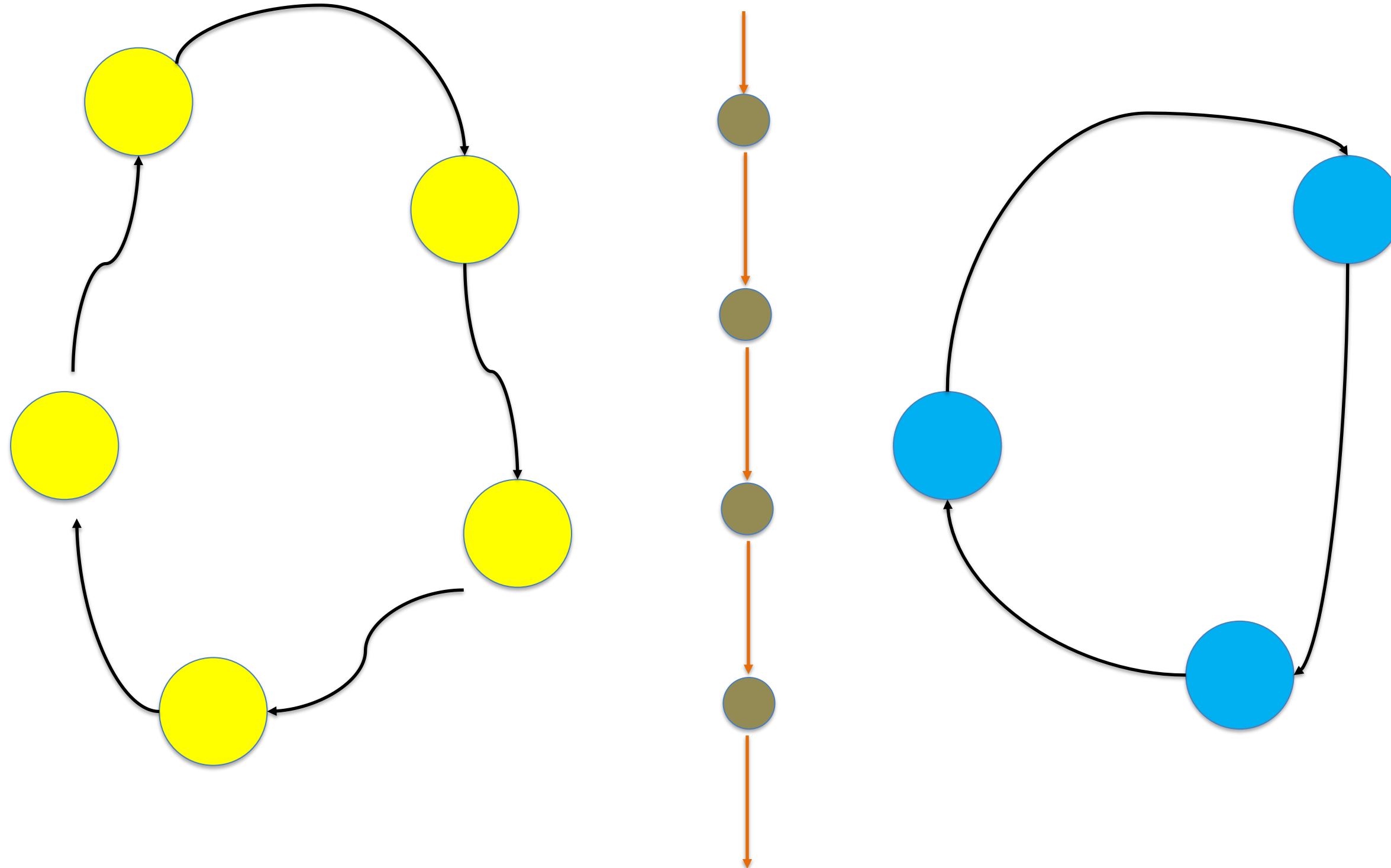


MODELLING protocols: Finite State Machines

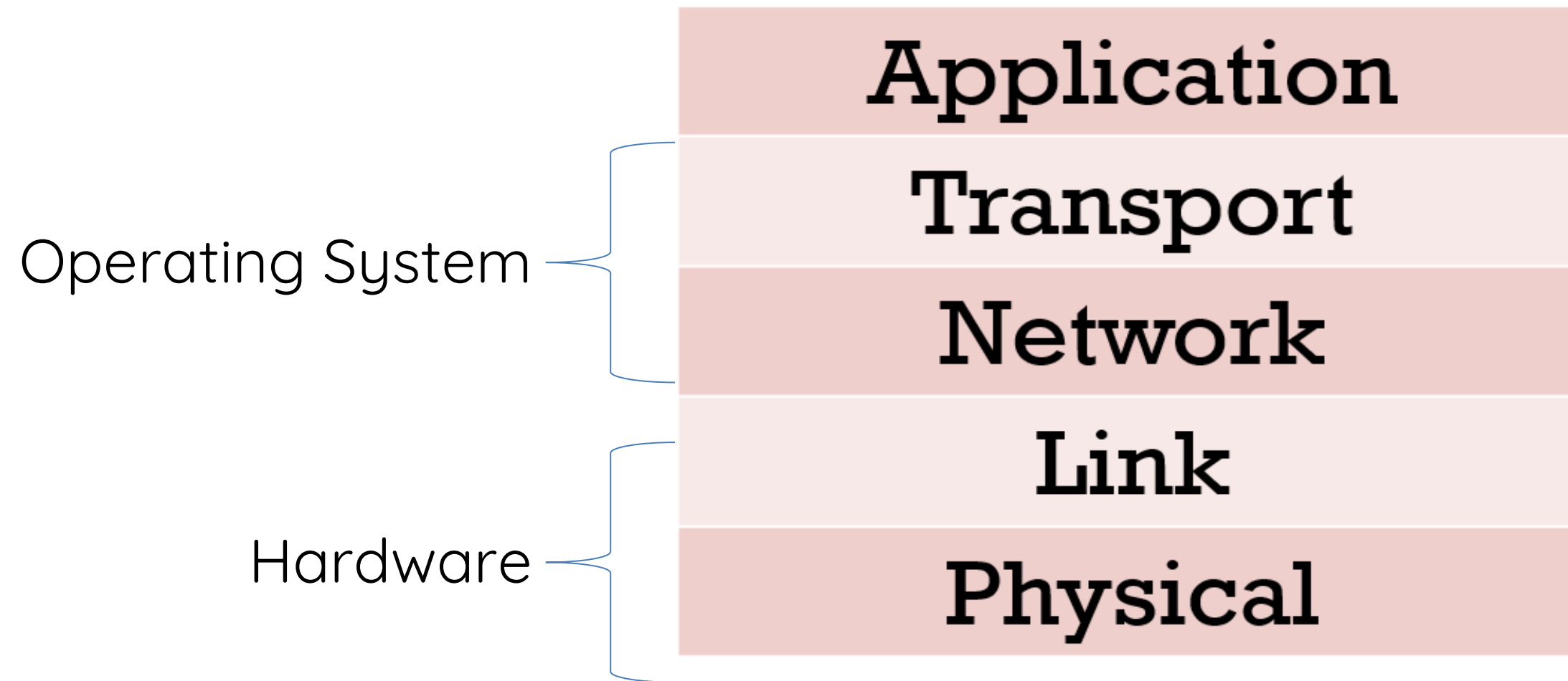


Mail

LINKED FINITE STATE MACHINES



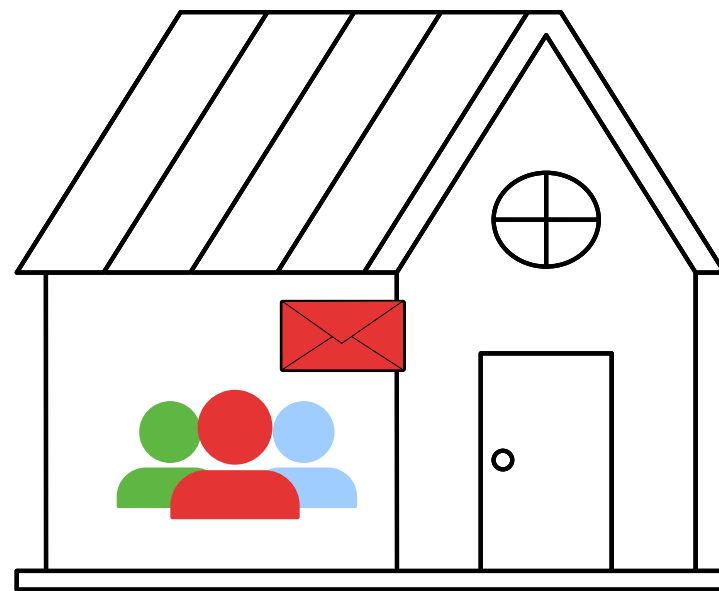
Protocol Stack



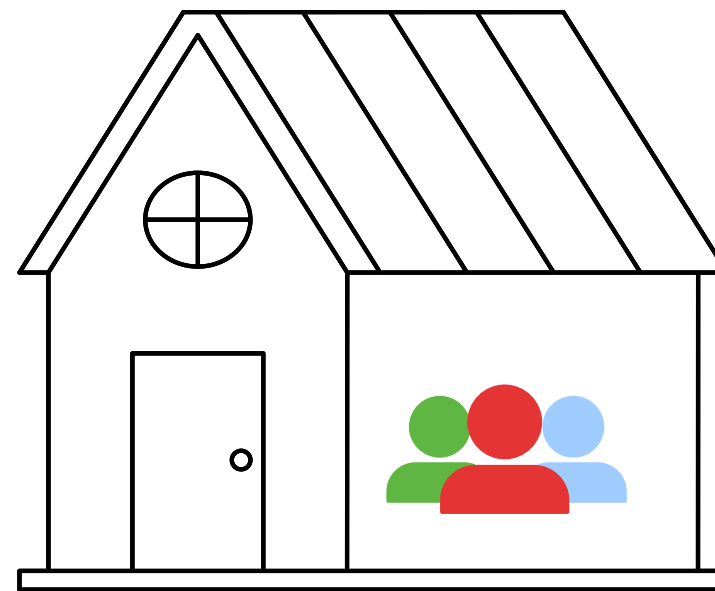
Daily Life Analogy

Application Layer Protocol

HTTP (Web), Email,
File Transfer,
Multimedia, etc.

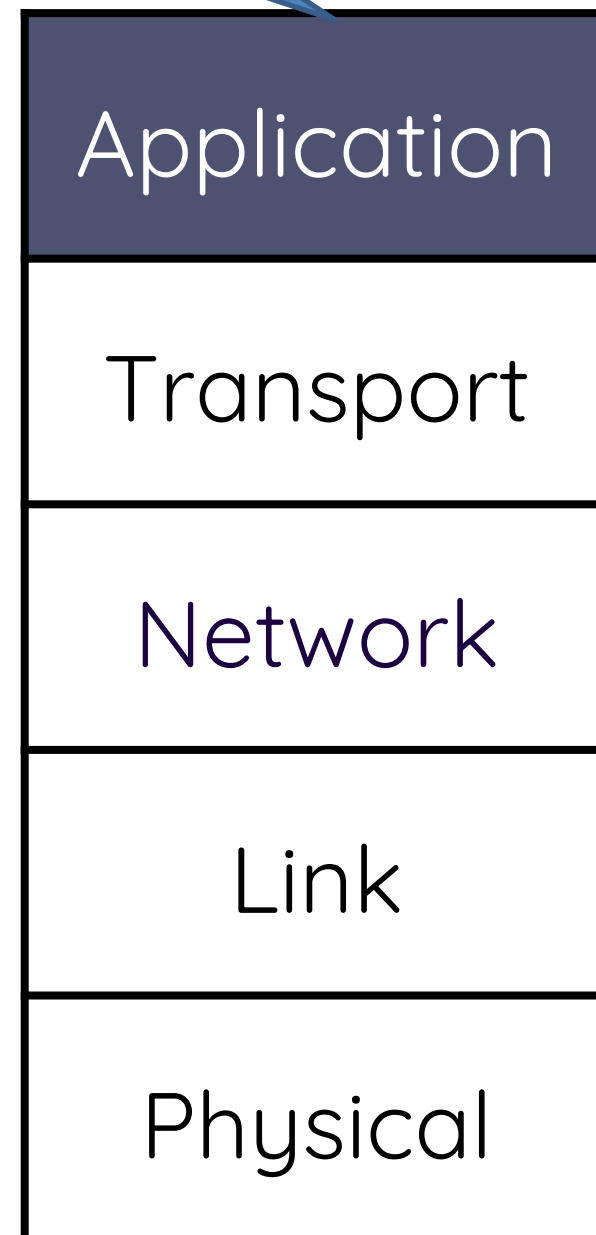


Process

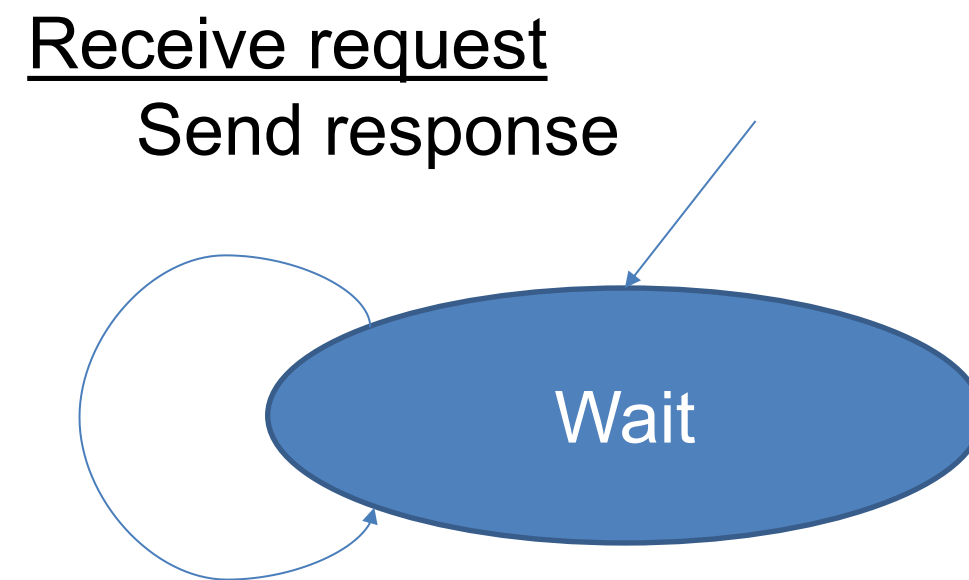


Process

Writing a letter using the proper
format expected by the recipient.

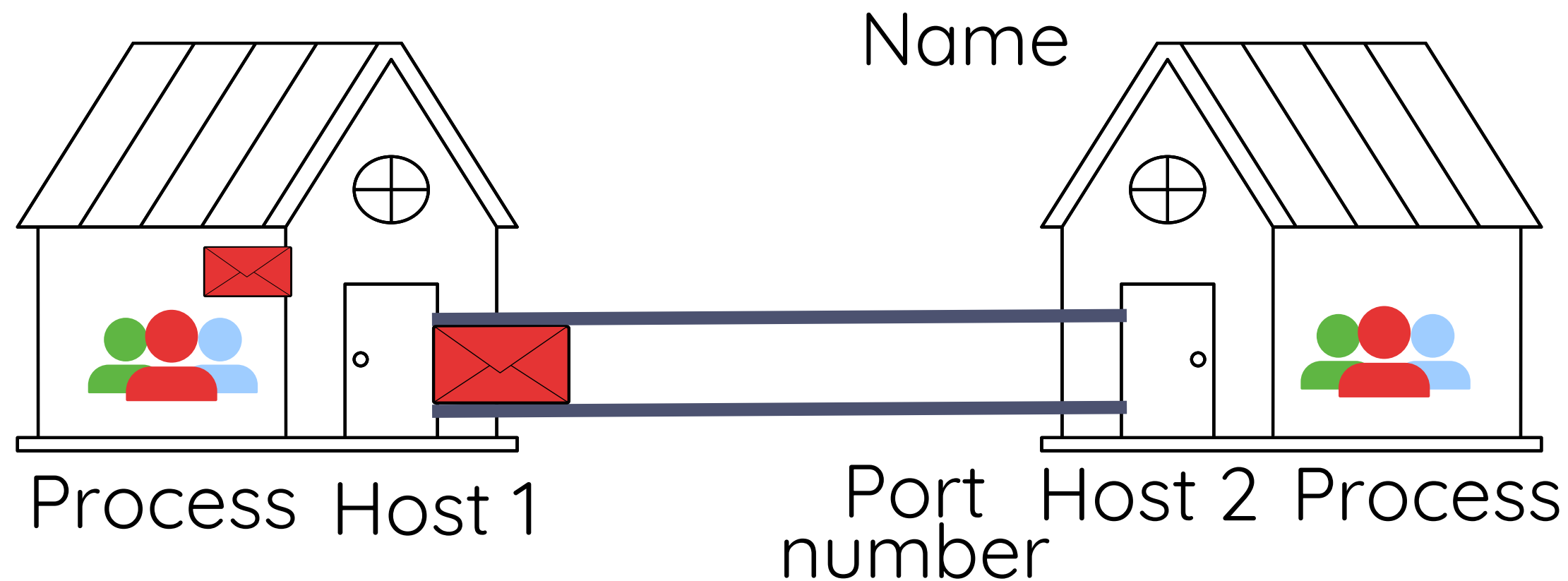


Web Server State Machine

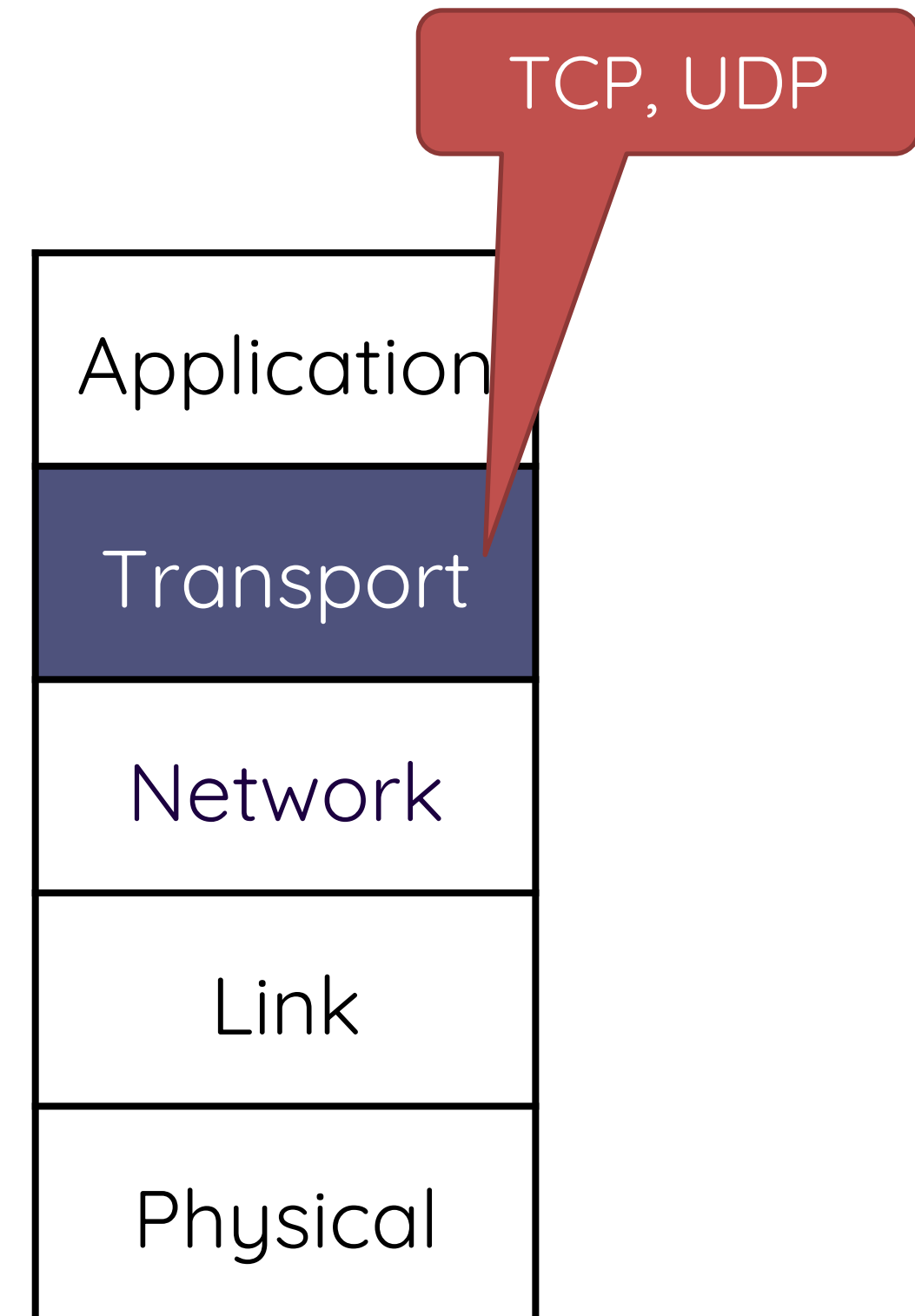


Daily Life Analogy

Transport Layer Protocol



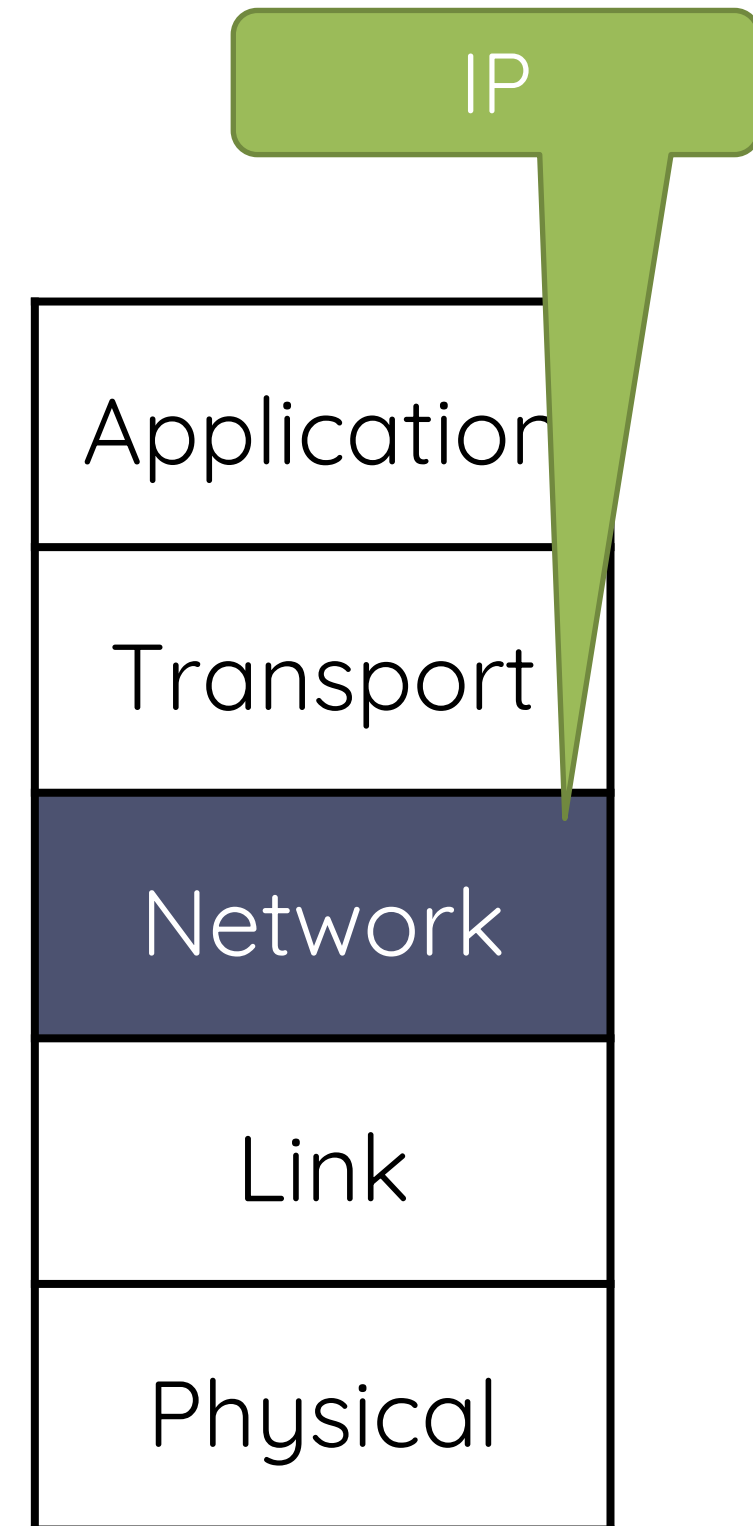
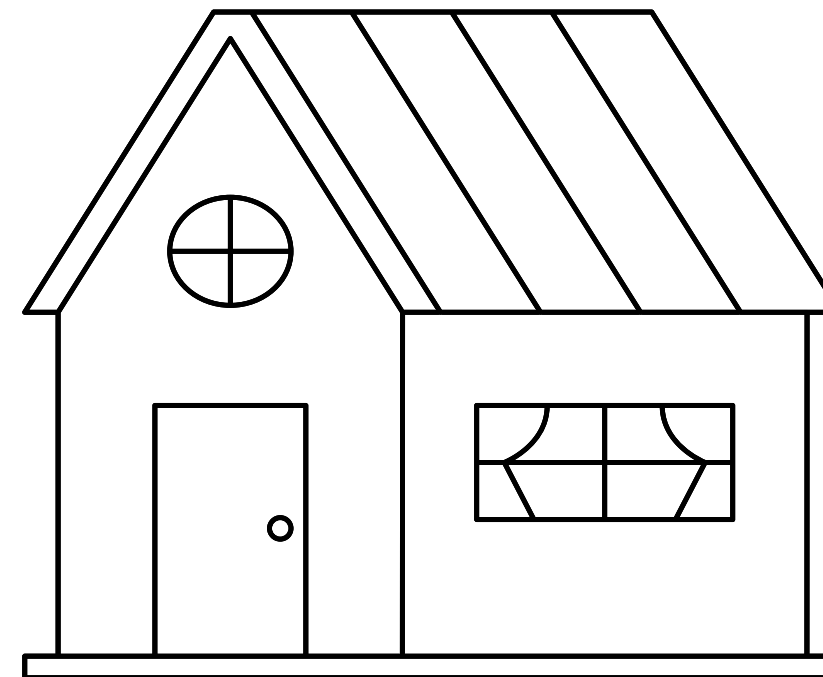
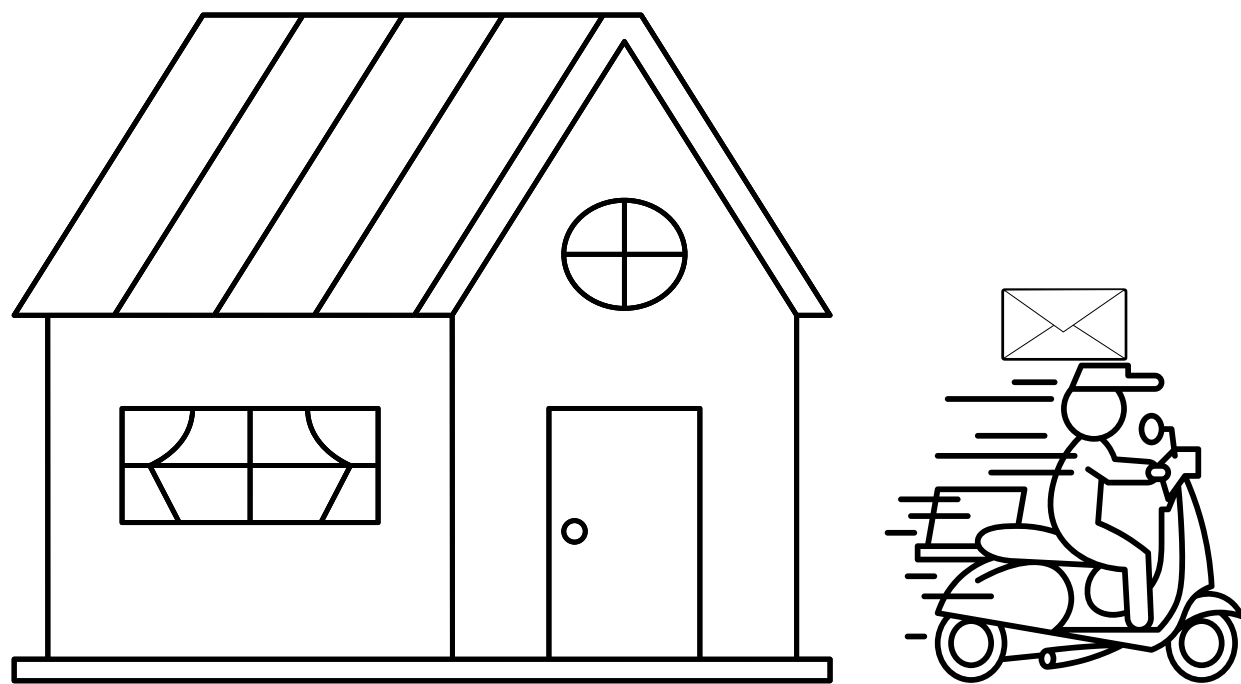
Handling reliability and making sure the message reaches the correct person.



Daily Life Analogy

Network Layer Protocol

IP Address

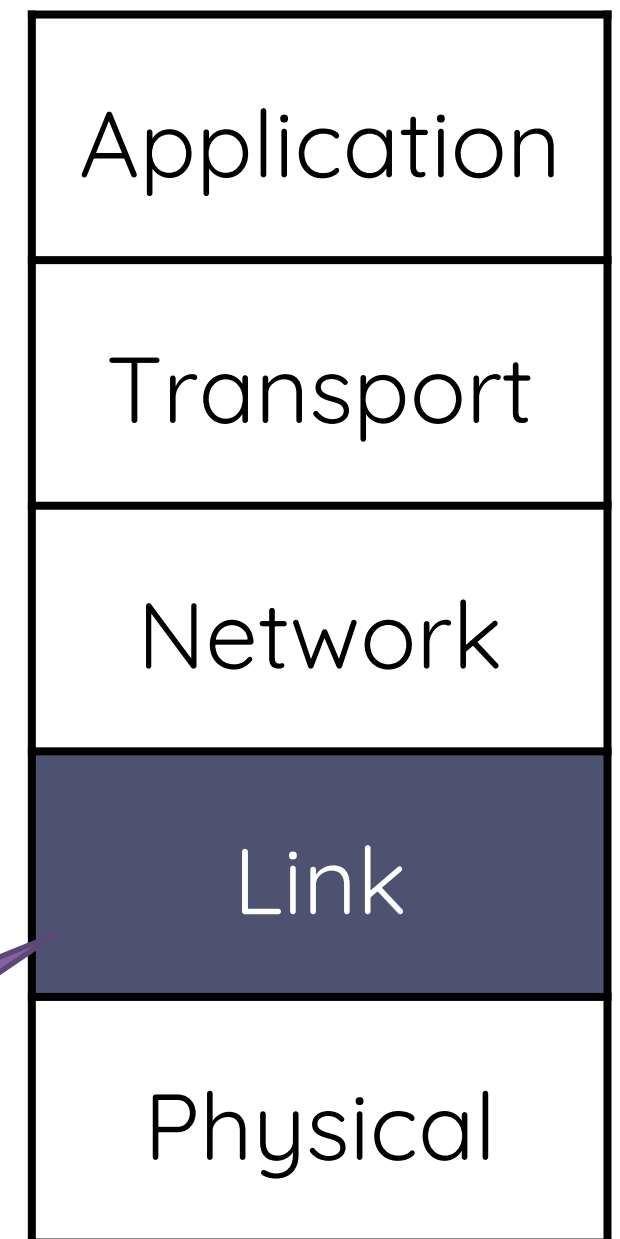


Like a routing officer at a post office or logistics center who decides the best sequence of roads, cities, or hubs to move the letter toward its goal.

Daily Life Analogy

Link Layer Protocol

Mac Address



Daily Life Analogy

Physical Layer Protocol

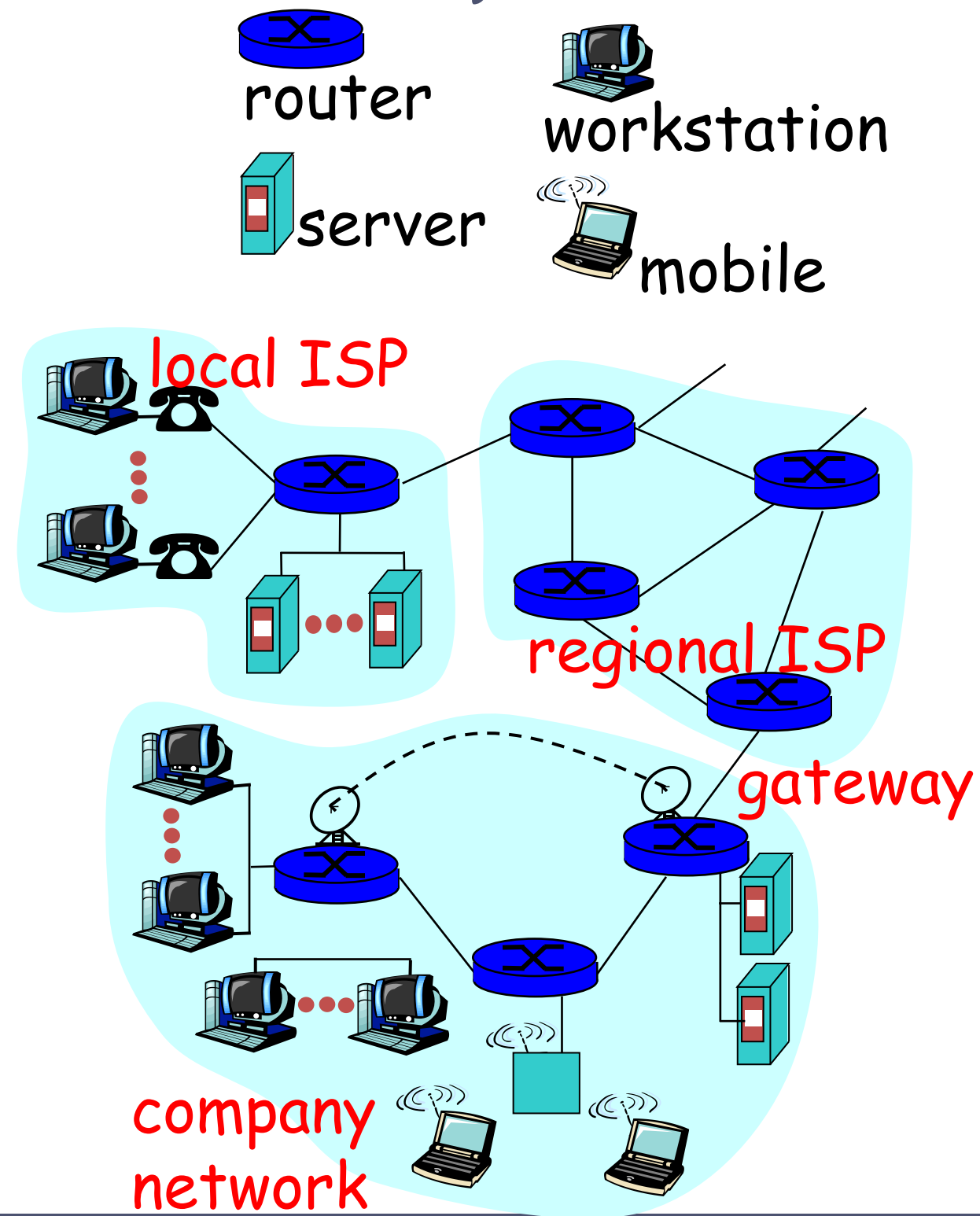


Make sure the letter is in the right container suitable for transport medium.

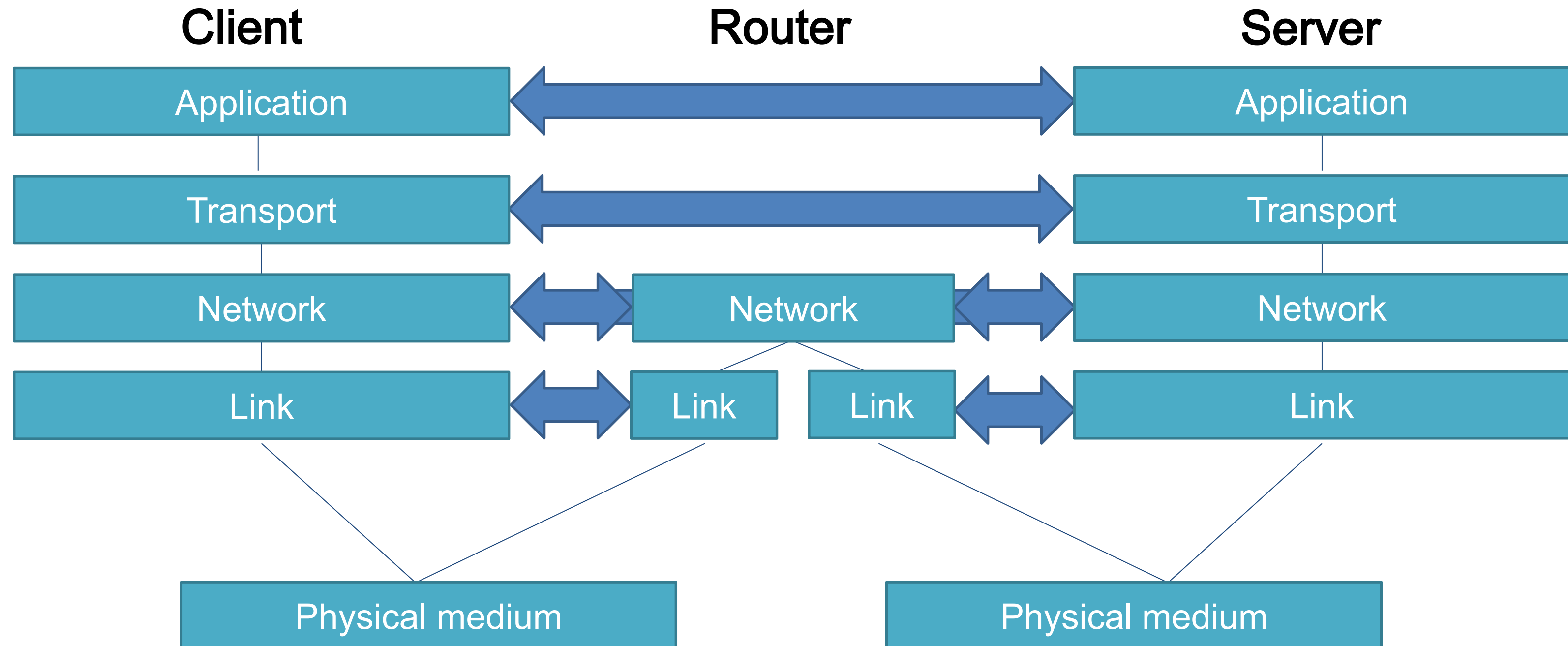
Sending Data through the Stack



As a network of networks, the internet has many routers

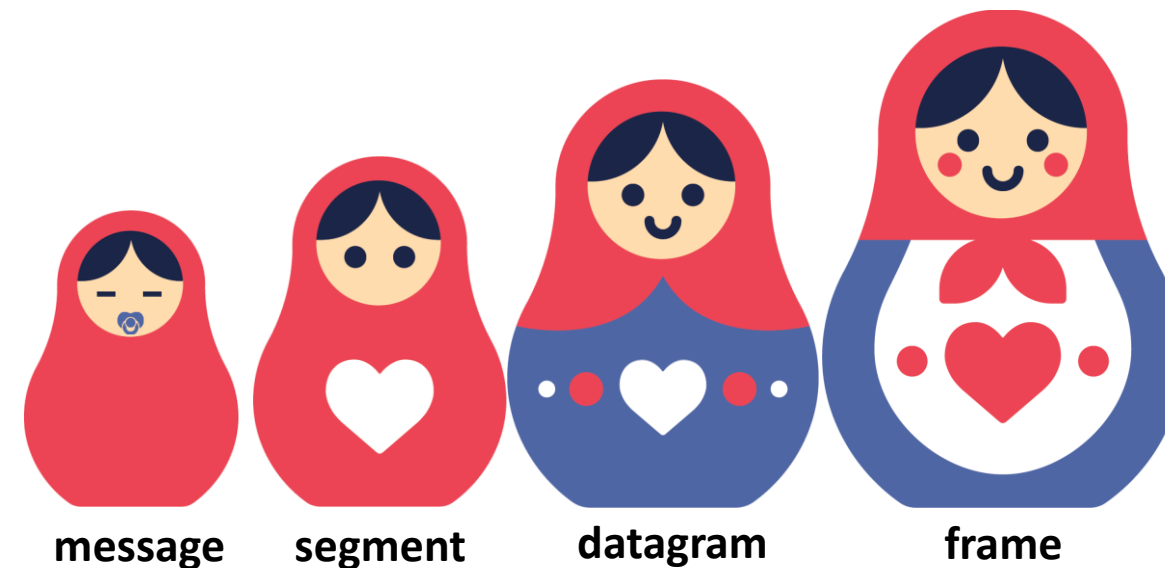


Sending Data through the Stack



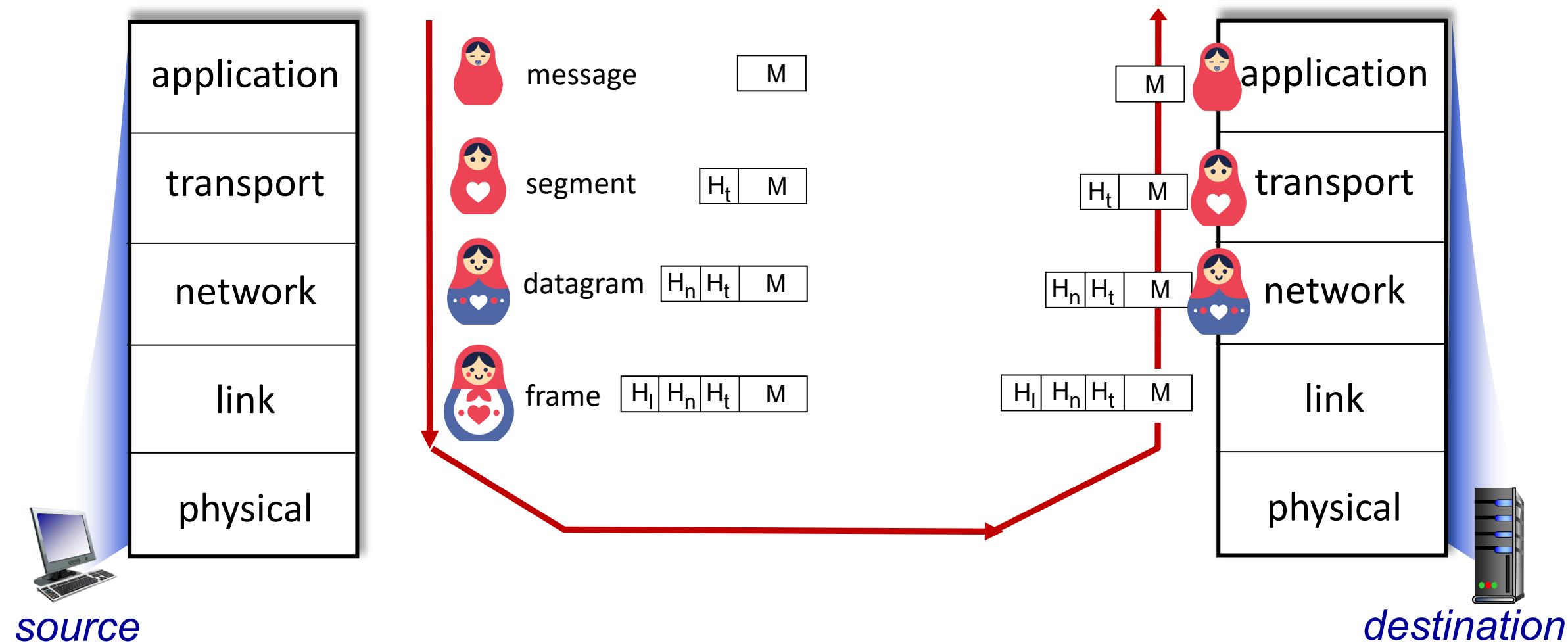
Encapsulation

Matryoshka dolls (stacking dolls)

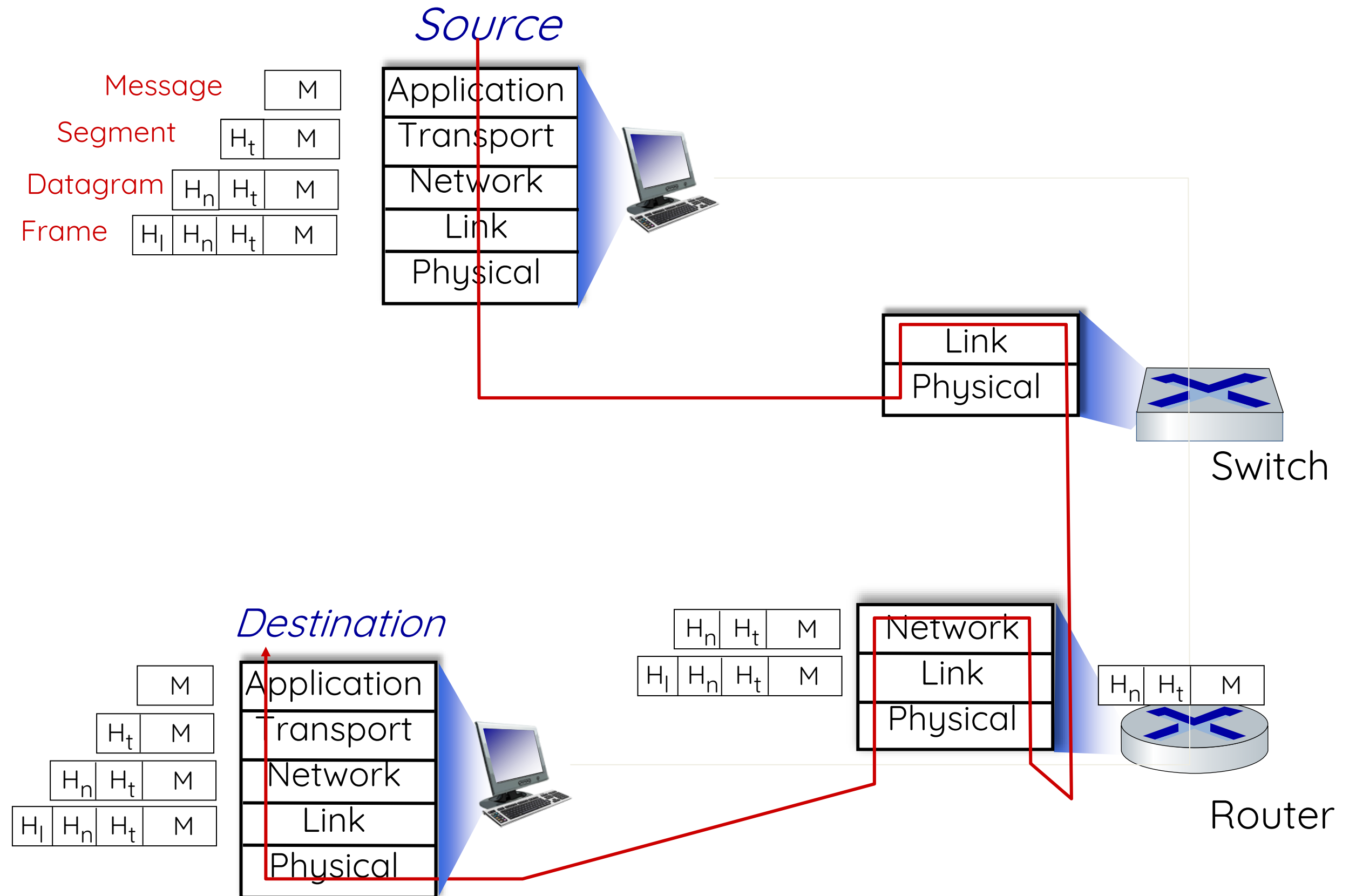


Protocol Layering and Data

- Each layer takes data from above
 - Adds header information to create new data unit
 - Passes new data unit to layer below



Encapsulation



Protocol Stack: Responsibilities

- **Transport layer:**
 - Identifies process on machine
 - Maybe resource within process (e.g., browser tab)
 - Ensures data arrives in order (if required)
 - Recovers lost data (if required)
- **Network layer:**
 - Routes datagrams through routers to destination machine
- **Link layer:**
 - Routes frames to adjacent machines (“direct” connection)
- **Physical layer:**
 - Encodes data appropriately for the physical medium

Summary of Module 1

- The Internet is a network of networks
 - End systems and routers
- How is data sent?
 - Data is chopped into packets
 - Each packet has its destination address
- Protocol stack
 - Each layer is responsible for a function
 - A protocol layer, in an abstract way, “talks” to a corresponding layer in another machine at the same level
 - Each layer requests services from the layer below it

In-class activity

- Form yourselves into groups (1 – 9 students per group)
- You should be able to chat conveniently in your group
- Go to PrairieLearn
- Click on Assessments
- Start the ICA12 assessment (Protocol Layers)
- Talk in your group about the answers
 - Hearing other students ideas
 - Explaining your ideas to others

Next Topic: Internet Performance